The Method of Recycled Predictions for Understanding Marginal Effects

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Motivation

- Say we are interested in the marginal effect of urban vs rural residence on the probability that a person identifies as being ‘happy’ or ‘not happy’.
- That is, we would like to estimate the probability of being happy that can be entirely ascribed to location of residence and, furthermore, to test whether these differ significantly between the groups.
- We, of course, control for covariates such as gender, age, income, education, marital status, and self-identified health rating.
- We could use the logit estimates from a multiple logistic regression model to predict the probability of being happy for each person in our data set.

Motivation

- The mean p, defined over all the N persons in the sample, will be the same as the sample proportion of persons that said they were happy.
- Similarly, the mean of p_{rural} defined over the M rural persons, and the p_{urban} defined over the N - M urban persons will be the same as the proportion of persons from each of these two groups that said they were happy.
- However, the difference between the two sample mean probabilities does not reflect the differences due solely to group membership. This is because persons from the two groups differ not just in terms of group identity but also with respect to variables like *gender, age, income, education, marital status, and health rating*.
- Computing the mean probabilities over each subgroup will not neutralize these differences, and hence, differences between p_{rural} and p_{urban} cannot be attributed to differences in group membership.

Motivation

- A standard approach is to set the other variables in the model equal to their means.
- However, recall that for non-linear functions, $E(X^k) \neq [E(X)]^k$.
- Further, urban persons may tend to vary from the mean of the covariates in a different direction or magnitude than rural persons.
- It's also problematic for categorical covariates.
- Hence, we consider the Method of Recycled Predictions to estimate the marginal effect of rural vs urban.

Method of Recycled Predictions

- The method of “recycled predictions” isolates the group effect on the predicted probability of being happy.
- First, we “pretend” that all N persons in the sample are from the rural group and holding the values of the other variables constant at their observed sample values, we then compute the probability for each individual and call it p_{rural_i}.
- Next, “pretend” that all N persons in the sample are from the urban group and, again holding the values of the other variables constant at their observed values, compute the probability for each individual and call it p_{urban_i}.

The Method of Recycle Predictions

- Since the values of the non-group variables are unchanged between these two hypothetical scenarios, the only difference between them is that, in the first scenario, the rural group coefficient is “switched on” (with the urban group coefficient “switched off”), while, in the other scenario, the urban coefficient is “switched on” (with the rural group coefficient “switched off”), for all the N persons in the sample.
- Consequently, the mean difference between p_{rural} and p_{urban} is entirely due to differences in group membership.
- The estimated probabilities can then be used in hypothesis testing and estimation.

Example

- We want to know the marginal effect of having a cesarian section on the risk of brachial plexus birth injury (BPBI).
- BPBI is rare but can cause permanent disability in the infant.
- We are controlling for previous BPBI event, number of previous children, mother's age at time of birth, mother's ethnicity, and mother's race.
- We fit the full model and save the parameters in a SAS data file.
- We then get the probabilities setting all observations to vaginal deliveries. Then set all observations to c-sections and again obtain the individual probabilities.
- We then conduct hypothesis testing on the probabilities.

SAS Code

```
/* Getting the parameters */
title " ";
proc logistic data = parityGE1 outmodel=parms_pbpi;
class bpbi csect (ref='1') mrace (ref="White") mhisp (ref='Non-Hispanic')
age(ref='1') prev_bpiYN (ref='0') / param=ref;
model bpbi (event='1') = csect prev_bpbiYN prev_kid age mhisp mrace
/parmlabel lackfit rsquare ctable pprob=(0.05 to 0.5 by 0.05) RIDGING=NONE;
run;
```

SAS Code

```
/* Getting probabilities pretending all had vaginal delivery */
data vag;
set parityGE1(rename=(csect=c_sect));
csect=0;
run;
****Apply model to entire data, assuming all vaginal;
PROC LOGISTIC inmodel=parms_pbpi Descending;
score data=vag
OUT=predVag(keep=hst id c_sect csect live_bpi p_1);
title "Applying the Risk Model to Entire Dataset Assuming all Patients had
Vaginal";
RUN;
```

SAS Code

```
/* Pretending all had c-section */
data ces;
set parityGE1(rename=(csect=c_sect));
csect=1;
run;
****Apply model to entire data, assuming all c-section ;
PROC LOGISTIC inmodel=parms_pbpi Descending;
score data=ces
OUT=predCes(keep=hst id c_sect csect live_bpi p_1);
title "Applying the Risk Model to Entire Dataset Assuming all Patients had
C-Section";
RUN;
```

SAS Code

```
/* Renaming probs so both datasets don't have same variable name */
data predvag;
set predvag;
p_vag = p_1;
drop p_1;
run;
proc sort data=predvag;
by hst id;
run;

data predces;
set predces;
p_ces = p_1;
drop p_1;
run;
proc sort data=predces;
by hst id;
run;
```

SAS Code

```
/* merging predicted probability datasets */
data preds;
merge predvag predces;
by hst id;
run;

title "Raw Incidence";
proc freq data=preds;
table c_sect*live_bpi/chisq relrisk;
run;

title "Test for differences in Pred Prob";
proc ttest data=preds;
paired p_vag*p_ces;
run;
```

SAS Code

```
title "Average Predicted Probability";
proc means data=preds;
var p_vag p_ces;
run;

title "Raw Incidence";
proc freq data=preds;
table c_sect*live_bpi/chisq relrisk;
run;

title "Test for differences in Pred Prob";
proc ttest data=preds;
paired p_vag*p_ces;
run;
```

SAS Output

- Table of Means

Variable	N	Mean	Std Dev	Minimum	Maximum
p_vag	2621705	0.0014684	0.000946971	0.000468407	0.0503408
p_ces	2621705	0.000188897	0.000124352	0.000060170	0.0067606

Paired t Test

N	Mean	Std Dev	Std Err	Minimum	Maximum
2621705	0.00128	0.000823	5.081E-7	0.000408	0.0436

DF	t Value	Pr > |t|
2.62E6	2518.48	<.0001

- Odds Ratio

Group	pred prob	1- pred prob	odds	OR
Vaginal	0.00147	0.99853	0.001472	7.787756
C-section	0.000189	0.999811	0.000189	

The Absolute Risk Reduction for c-section is 0.00128.
Hence, the number of c-sections needed to prevent one, BPBI is
NNP = 1/0.00128 = 781